

BTEC Level 3 Applied Science • Unit 1 Chemistry • Answer sheet

Ionic Bonding — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording.

A

1. Electrostatic attraction between oppositely charged ions.
2. A metal and a non-metal.
3. Ions are not free to move.

B

4. Na loses 1 electron to form Na^+ . Cl gains that electron to form Cl^- . Oppositely charged ions attract.
5. Strong attractions throughout a giant lattice need a lot of energy to overcome.
6. Ions become free to move and carry charge.

C

7. Na_2O ; MgCl_2 ; Al_2O_3 .
8. A shift lines up like charges, which repel, so the crystal splits.
9. Molten NaCl : mobile ions. Copper: delocalised electrons.

D

10. Mg loses $2e^- \rightarrow \text{Mg}^{2+}$. O gains $2e^- \rightarrow \text{O}^{2-}$. Electrostatic attraction. Giant lattice of Mg^{2+} and O^{2-} .
11. Strong ionic attractions throughout the lattice $\rightarrow$ high m.p. Solid ions fixed $\rightarrow$ no conduction.
12. $\text{Al}_2(\text{SO}_4)_3$ so +6 balances -6.

E

13. Mg^{2+} and O^{2-} have higher charges than Na^+ and Cl^- , so the electrostatic attraction is stronger and more energy is needed to melt the lattice.